

Spatial clustering of amyotrophic lateral sclerosis and the potential role of BMAA.

Amyotrophic lateral sclerosis (ALS) is a fatal neurodegenerative syndrome which has no known cause, except for a small proportion of cases which are genetically inherited. The development of ALS likely involves both genetic and environmental risk factors. Environmental risk factors implicated in ALS have included heavy metals, trauma, pesticides, electrical injuries, electromagnetic radiation and the cyanobacterial-derived neurotoxin beta-N-methylamino-L-alanine (BMAA). To investigate possible environmental risks, a number of epidemiological studies of ALS have been conducted. Some of these studies employ spatial analysis techniques that examine for spatial clusters of ALS and can help guide further research into identifying environmental exposures. Despite identifying geographical disparities in the distribution of ALS cases, these studies have not provided any clear associations with environmental factors. We review the literature on important studies of spatial clustering of ALS and explore the hypothesized link between the neurotoxin BMAA and ALS.